9. option(a)

In zero order reactions, increase in the concentration of reactant does not alter the rate. So statement (i) is wrong.

$k = A\, e^{-\left(\frac{E_a}{RT}\right)}$

if $E_a = 0$ so, statement (ii) is correct, and statement (iii) is wrong

$k = A\, e^{0}$

$k = A$

$\ln k = \ln A - \left(\frac{E_a}{R}\right)\left(\frac{1}{T}\right)$

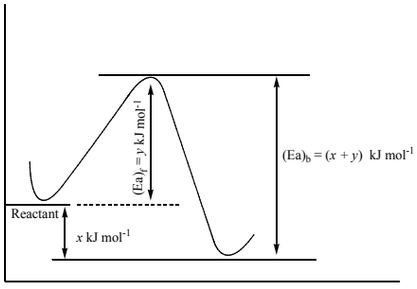
this equation is in the form of a straight line equation

$y = c + m\, x$

a plot of $\ln k$ vs $\left(\frac{1}{T}\right)$ is a straight line with negative slope

so statements (iv) and (v) are wrong.

10. option(d)



$(x+y)\,\text{kJmol}^{-1}$

$(x+y)\times 10^{3}\ \text{Jmol}^{-1}$

11. option(c)

$T_1 = 200K\ ;\ k = k_1$

$T_2 = 400K\ ;\ k = k_2 = 2k_1$

$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1\ T_2}\right)$

$\log\left(\frac{2\,k_1}{k_1}\right) = \frac{E_a}{2.303 \times 8.314\ \text{J K}^{-1}\text{mol}^{-1}}\left(\frac{400\ \text{K} - 200\text{K}}{200\text{K} \times 400\text{K}}\right)$

$E_a = \frac{0.3010 \times 2.303 \times 8.314\ \text{JK}^{-1}\,\text{mol}^{-1} \times 200\ \text{K} \times 400\ \text{K}}{200\ \text{K}}$

$E_a = 2305\ \text{J mol}^{-1}$

$E_a = 2.305\ \text{kJ mol}^{-1}$

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